# Part B Labs: Fine-Tuning, Evaluation & Integration

**Main (use first — applies to everyone)**

- PyTorch reproducibility / randomness: https://pytorch.org/docs/stable/notes/randomness.html
- TensorFlow determinism: https://www.tensorflow.org/api_docs/python/tf/config/experimental/enable_op_determinism
- Python requirements files (pip): https://pip.pypa.io/en/stable/user_guide/#requirements-files
- Model evaluation basics (scikit-learn): https://scikit-learn.org/stable/modules/model_evaluation.html
- Confusion matrix (scikit-learn): https://scikit-learn.org/stable/modules/generated/sklearn.metrics.confusion_matrix.html
- MLflow (experiment tracking): https://mlflow.org/docs/latest/index.html
- FastAPI quickstart: https://fastapi.tiangolo.com/tutorial/first-steps/
- Python logging: https://docs.python.org/3/library/logging.html
- pytest: https://docs.pytest.org/en/stable/
- Docker getting started (optional): https://docs.docker.com/get-started/
- TorchMetrics: https://lightning.ai/docs/torchmetrics/stable/

---

**Airport Security (Machine Vision track)**

- PyTorch Transfer Learning tutorial: https://docs.pytorch.org/tutorials/beginner/transfer_learning_tutorial.html
- TorchVision detection finetuning tutorial: https://pytorch.org/tutorials/intermediate/torchvision_tutorial.html
- COCO detection evaluation (mAP/IoU): https://cocodataset.org/#detection-eval

---

**LLM Dialogue System (Customer Support track)**

- Hugging Face Transformers training: https://huggingface.co/docs/transformers/training
- Hugging Face Trainer API: https://huggingface.co/docs/transformers/main_classes/trainer
- PEFT (LoRA / parameter-efficient fine-tuning): https://huggingface.co/docs/peft/index

- MultiWOZ repository: https://github.com/budzianowski/multiwoz
- Bitext customer support dataset (Hugging Face): https://huggingface.co/datasets/bitext/Bitext-customer-support-llm-chatbot-training-dataset

---

**Most interactive**

**Main (interactive for everyone)**
• Kaggle Learn (interactive micro-courses): https://www.kaggle.com/learn
• Google Colab (run notebooks in the browser): https://colab.research.google.com/
• fast.ai course (notebook-driven, very hands-on): https://course.fast.ai/
• Learn PyTorch (interactive-style online book + exercises): https://www.learnpytorch.io/
• W3Schools (Python quick refresher): https://www.w3schools.com/python/
• W3Schools (NumPy basics): https://www.w3schools.com/python/numpy/default.asp
• W3Schools (Python Machine Learning – intro): https://www.w3schools.com/python/python_ml_getting_started.asp

**Airport Security (Machine Vision — interactive)**
• Kaggle Learn: Intro to Deep Learning: https://www.kaggle.com/learn/intro-to-deep-learning
• Kaggle Learn: Computer Vision: https://www.kaggle.com/learn/computer-vision
• PyTorch 60-Minute Blitz (hands-on): https://docs.pytorch.org/tutorials/beginner/deep_learning_60min_blitz.html

**LLM Dialogue (Customer Support — interactive)**
• Hugging Face LLM Course (hands-on chapters): https://huggingface.co/learn/llm-course/en/chapter1/1
• W3Schools GenAI (Prompting overview / background): https://www.w3schools.com/aws/genai/aws_prompt_prompting.php

---